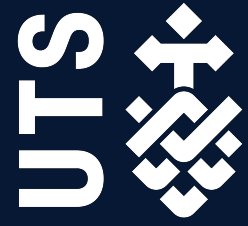


Assignment 2:

Data Analysis Report



36103: Statistical Thinking for Data Science

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Group Number: 5

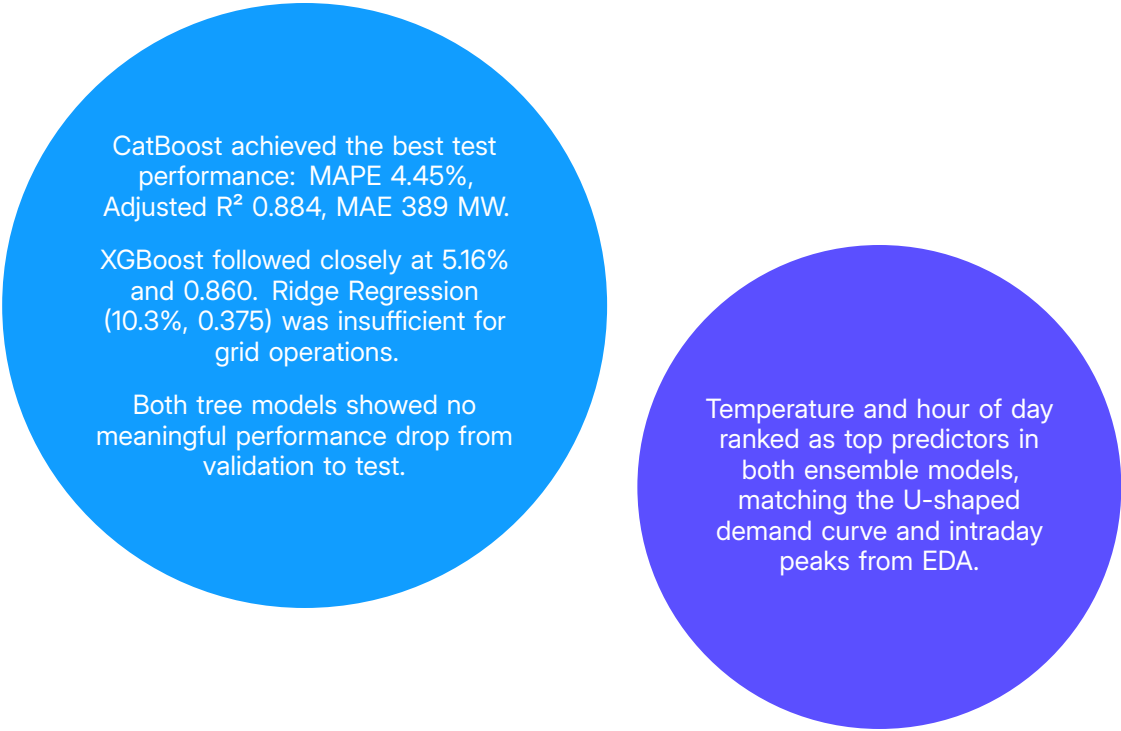
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Executive Summary

Electricity demand in NSW changes with weather, time of day, and season. During peak periods, wholesale prices can reach \$17,500/MWh, so reliable short-term forecasts matter for both cost management and grid stability.

This project used high-frequency NSW data from 2022-2026 to study demand patterns and build forecasting models. Exploratory analysis showed demand is lowest in mild conditions, rises during extreme hot or cold weather, and differs between weekdays and weekends. Rooftop solar reduces midday demand but causes a sharp evening increase.

CatBoost is the recommended model. Its average error of around 389 MW means forecasts should inform decisions, not drive them outright, particularly during extreme weather or supply disruptions.



CatBoost achieved the best test performance: MAPE 4.45%, Adjusted R^2 0.884, MAE 389 MW.

XGBoost followed closely at 5.16% and 0.860. Ridge Regression (10.3%, 0.375) was insufficient for grid operations.

Both tree models showed no meaningful performance drop from validation to test.

Temperature and hour of day ranked as top predictors in both ensemble models, matching the U-shaped demand curve and intraday peaks from EDA.

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Introduction

1 Introduction

1.1 Problem Statement

Electricity supply and demand must balance every 30 minutes. When they don't, grid operators absorb the cost. NSW wholesale prices can reach \$17,500/MWh under the market cap that applied through mid-2025 (AER, 2025c). Accurate short-term demand forecasting keeps dispatch decisions ahead of those spikes.

1.2 Rationale

The National Electricity Market (NEM) serves roughly 11 million customers across eastern Australia, with NSW being the country's largest electricity generating state (AER, 2026). Demand shifts every 30 minutes with temperature, time of day, and day type, and the grid has no meaningful storage to absorb mismatches.

The supply side is also changing. Australia ranks first globally in per capita solar electricity generation (Green.com.au, 2026), with NSW holding the highest installed rooftop capacity of any state at 7.5 GW as of mid-2025 (CEC, 2025). Rooftop solar suppresses midday grid demand but drops off sharply in late afternoon, creating a steep evening ramp that dispatchable generators must cover at short notice. NSW already sees a 34-41% gap between its daily minimum and maximum demand, among the largest swings in the NEM (AER, 2025b).

Prices reflect that volatility. In 2024, NSW average wholesale electricity prices rose 43% year-on-year to \$150/MWh (AER, 2025a). Forecasting errors at that price level carry direct costs for operators, retailers, and consumers.

1.3 Project Aims and Objectives

The aim of this project was to determine whether machine learning models trained on historical half-hourly NSW data can predict grid electricity demand accurately enough to support operational dispatch planning. To support this aim, we asked:

1. Does temperature have a non-linear relationship with NSW grid demand across 2°C bins from 5°C to 43°C?
2. Do mean hourly demand profiles differ between weekdays, weekends, and public holidays in NSW?
3. Can an ensemble regression model predict 30-minute NSW grid demand with a MAPE below 5% on held-out test data?

To answer these questions, we sourced and merged half-hourly AEMO data with weather, rooftop solar, and economic indicators, then ran exploratory analysis to identify key demand drivers. We built and compared Ridge Regression, XGBoost, and CatBoost models on a 60:20:20 chronological split, evaluating each on RMSE, MAE, MAPE, and Adjusted R².

Methodology

2 Methodology

2.1 Models

2.1.1 Linear Regression (Ridge)

The linear regressor is a parametric model that fits a linear equation over observational data between a target feature and multiple independent predictors. It served as the baseline noting its computational efficiency and simplicity in capturing non-linear demand patterns. The Ridge variant was chosen to provide a stable baseline by focusing on L2 regularisation to mitigate multicollinearity among highly correlated weather features.

2.1.2 Gradient Boosting Decision Trees

Gradient Boosted Decision Trees (GBDT) are a non-parametric ensemble learning technique that sequentially builds decision trees, where each successive learner is optimised via gradient descent to minimise the residual loss of the previous ensemble.

Table 1. Model Selection and Theoretical Rationale

Model	Definition	Methodology Rationale
XGBoost	Scalable GBDT library designed for high computational efficiency.	Selected to address non-linear relationships in the data through sequential tree-building and dual regularisation (L1 and L2) to control model complexity.
CatBoost	Advanced GBDT algorithm that utilises symmetric trees and ordered boosting to reduce prediction shift and handle categorical features.	Chosen for its superior ability to generalise on unseen data and its efficient handling of categorical features (such as 'Season' and 'Holiday') without extensive manual encoding.

2.2 Grid Search

Hyperparameter optimisation is performed via Grid Search with k -fold Cross-Validation. Through exhaustive predefined parameter grids, we identified configurations that minimised generalisation error of each model.

2.3 Metrics

Model performance was evaluated using a hold-out test set to ensure generalisability, with predictive accuracy quantified through Mean Absolute Percentage Error (MAPE) and Root Mean Square Error (RMSE) to penalise significant forecasting deviations during peak periods. Adjusted R^2 was utilised to measure the proportion of variance explained while accounting for the increased complexity of the multi-feature feature set. The following Table 2 below illustrates the chosen metrics.

Table 2. Model Performance Metrics

Metric	Purpose	Formula
MAE	Average absolute error in MW; operationally interpretable.	$\frac{1}{n} \sum y_i - \hat{y}_i $
MAPE	Scale-independent accuracy across demand magnitudes.	$\frac{1}{n} \sum \left \frac{y_i - \hat{y}_i}{y_i} \right \times 100$
RMSE	Penalises large outliers in peak demand periods.	$\sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$
Adjusted R ²	Explained variance penalised for predictor count.	$1 - \frac{(1-R^2)(n-1)}{n-p-1}$

Results

3 Results

Conclusion

4 Conclusion

4.1 Conclusion

4.2 Discussion

4.3 Project Limitations and Caveats

4.4 Stakeholder Analysis and Project Outcomes

References

5 References

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Appendices

6 Appendices